CHRISTOPHER EBUKA OJUKWU

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RESEARCH INTERESTS

Computational and statistical genetics and genomics; genetic architecture of complex traits; translational genomics; therapeutic target discovery.

EDUCATION

Ph.D., Computer Science | University of Colorado Boulder | Expected May 2027
Advisor: Daniel Acuña

M.S., Computer Science | University of Colorado Boulder | 2026

Postgraduate Diploma (PGD), Computer Science | Nile University of Nigeria | 2021
GPA: 5.0/5.0 (Graduated top of class)

Higher National Diploma (HND), Computer Science | Delta State Polytechnic, Nigeria | 2017

TECHNICAL SKILLS

Programming: Python, Bash
Statistical & Genetic Analysis: GWAS, SNP heritability estimation, colocalization, population structure analysis, cross-ancestry analyses, network-based genetic analysis, therapeutic target evaluation, drug development outcome analysis
Genomics Tools: PLINK/PLINK2, REGENIE, LDSC, MAGMA
Computing & Infrastructure: High-performance computing (HPC), Linux/UNIX, cloud computing (AWS), SLURM, computational pipeline design and development, workflow automation, large-scale biobank data analysis (UK Biobank, All of Us)
Version Control: Git, GitHub

RESEARCH EXPERIENCE

Graduate Researcher | Science of Science & Computational Discovery Lab 2024 - Present
Department of Computer Science, University of Colorado Boulder

- Analyzed 790 genetically supported drug target-indication pairs and found that broader ancestry representation did not independently predict drug launch after accounting for research volume, distinguishing the role of ancestry diversity in therapeutic development.
- Quantified cross-ancestry portability of GWAS associations, finding attenuated genetic effect estimates in non-European populations (portability slope = 0.68) and heterogenous causal signals across loci, highlighting limitations in the generalizability of European-derived genetic evidence.
- Investigating how scientific publications, research attention, and retracted evidence shape the observed structure and reliability of protein-protein interaction networks.

Graduate Researcher | Evans Lab, Institute for Behavioral Genetics 2026 - Present
University of Colorado Boulder

- Integrated GWAS associations with protein-protein interaction networks to investigate the genetic architecture of height and schizophrenia, finding that common-variant-associated genes occupy non-random positions relative to monogenic disease genes within a biological network.
- Developed large-scale height GWAS across UK Biobank, All of Us, and HRS to expand sample size and genomic variant coverage, addressing limitations in existing datasets and strengthening statistical power for investigating complex-trait genetic architecture.
- Developing a reusable computational pipeline that integrates GWAS with protein-protein interaction networks, enabling researchers to systematically characterize the genetic architecture of complex traits without rebuilding trait-specific analyses.

Graduate Researcher | Layer Lab 2022 - 2024
Department of Computer Science, University of Colorado Boulder

- Contributed to computational approaches for somatic structural variant detection and gene fusion identification by

RESEARCH PUBLICATIONS AND MANUSCRIPTS

Under Review

- **Ojukwu, C. E.**, Acuna, D. E. *How and When Ancestry Matters in Genetic Evidence for Drug Targets*. Under review at *PLOS Genetics*, 2026.
- Meguimtsop, A. D. M., **Ojukwu, C. E.**, Taechoyotin, P., Chavez-Ruelas, C., Burke, R., Clauset, A., and Acuna, D. E. *The Disparate Beliefs of Large Language Models About Science and Scientists*. Under review at the *Proceedings of the National Academy of Sciences*, 2026.

Research in Progress

- **Ojukwu, C. E.**, Acuna, D. E. *Research Attention and the Accumulation of Knowledge in Protein-Protein Interaction Networks*.
- **Ojukwu, C. E.**, Acuna, D. E. *Retraction-based Auditing of Protein-Protein Interaction Networks*.
- **Ojukwu, C. E.**, Evans, L. M. *The Network Architecture of Common-Variant Associations in Complex Traits*.

CONFERENCE PRESENTATIONS

- **Ojukwu, C. E.** *Ancestry Composition of Genetic Support for Drug Targets*. Oral presentation, *International Conference on the Science of Science and Innovation*, June 2026.
- **Ojukwu, C. E.** *A Novel Approach for Comprehensive Somatic Structural Variation Detection and Gene Fusion Identification*. Oral presentation, *Genome Informatics Conference*, Syosset, NY, December 2023.

TEACHING EXPERIENCE

Graduate Part-Time Instructor | University of Colorado Boulder, College of Engineering & Applied Science | Jun–Jul 2025

- Developed instructional materials and taught CSCI 3308: Software Development Methods and Tools.

Graduate Teaching Assistant | University of Colorado Boulder, College of Engineering & Applied Science | Aug 2022–Present

- Courses: CSCI 1300 Starting Computing; CSCI 3104 Algorithms; CSCI 4308/4318 Software Project Management and Mentoring.
- Managed undergraduate software development teams as part of CS Capstone Projects (CSCI 4308/4318), including a team that won first at the 2025 Engineering Projects Expo.

HONORS, FELLOWSHIPS & AWARDS

Summer Research Fellowship | University of Colorado Boulder, Department of Computer Science | 2026

- Awarded $7,500 summer research funding to support doctoral research, selected as one of seven recipients.

UK Study Tour Selection | British Council, Nigeria Creative Enterprise Support Program | 2019

- Selected as one of five participants for a UK study tour focused on entrepreneurship, creating enterprise development, and international collaboration.

ADDITIONAL PROFESSIONAL EXPERIENCE

Fashion Journalist / Contributor | The Guardian Nigeria, V Magazine, BellaNaija Style | 2017 –2020

- Authored published articles, interviews, and feature stories covering culture, fashion, and entrepreneurship.

Program Director | 360 Creative Hub | 2018 –2019

- Led program development and coordinated initiatives supporting emerging fashion businesses.